

# Unwrapping the Gift: Benefits of Computer Technology

## Lecture 2

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# Benefits to medicine



- ❑ Sophisticated medical imaging
- ❑ Reduced surgery time
- ❑ Training surgeons
- ❑ Faster recovery time (e.g. minimally invasive surgery)
- ❑ Patient monitoring
- ❑ Improved treatments
- ❑ Patient records
  - More legible
  - Better organized
  - Accessible by many sites, simultaneously
  - Available immediately

# Benefits to medicine



## ❏ Diagnosis

- Streamlined diagnosis
- Improved accuracy
- Reduced need for biopsy
- Improved screening and predictions

## ❏ Telemedicine

- Broad access to medical help
- Less expensive
- Expands consultations beyond rural/remote sites



# Benefits to disabled



- ❑ Disability-specific computer applications (e.g., a braille printer, large size print).
- ❑ Control of household and workplace appliances via speech (people who can't use their hands.)
- ❑ Mobility
- ❑ Control of artificial limbs
- ❑ Improved vision (e.g. large font sizes)
- ❑ Access to adaptive educational equipment
- ❑ Improved communication (e.g. speech input for people who can't write, speech synthesis for those who can't talk)
- ❑ Opportunity to return to work

# Benefits – Automation



- ❑ Organize and access large inventories
- ❑ Cost-saving
- ❑ Time-saving
- ❑ Improved customer satisfaction
- ❑ Tedious, monotonous work done by machines
- ❑ Improved workplace safety

# Identifying people and products



- ❑ E.g. Bar codes & smart cards –
- ❑ Time-saving
- ❑ Accuracy
- ❑ Ease-of-use
- ❑ Cost-saving
- ❑ More secure
- ❑ Customizable
- ❑ Reliable

# Monitoring Environment and Materials



- ❑ E.g. Bar codes & smart cards –
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- ❑ Accuracy
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# Benefits of Computers in Reducing Paper Use and Garbage



- ❑ Send/receive digital documents instead of hardcopy.
- ❑ Read, write, and edit online.
- ❑ Cost-saving; smaller storage needed.
- ❑ Creation of toxic wastes at paper mills reduced.

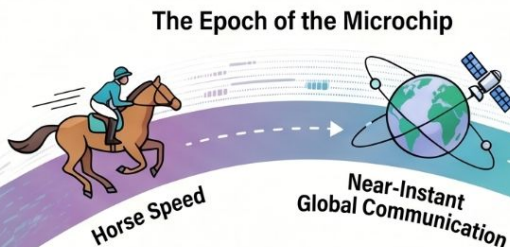
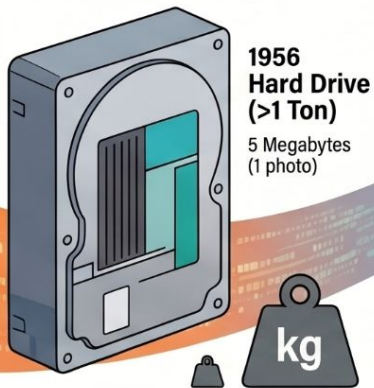


# Other Benefits

- ❑ Tracking
- ❑ Rare plants and animals
- ❑ Migration habits
- ❑ Stolen goods
- ❑ Lost pets or people

# Unwrapping the Gift: The Transformative Power of Computer Technology

## The Velocity of Progress



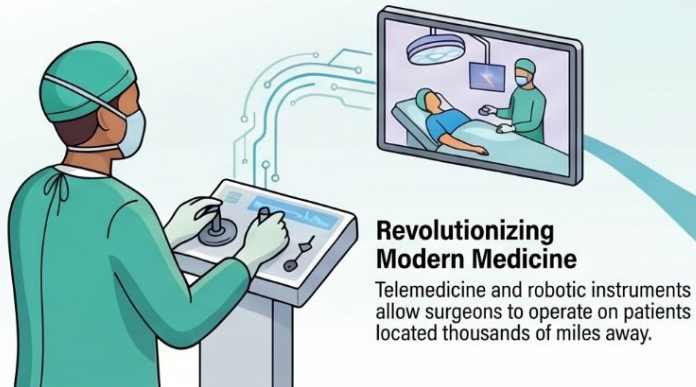
## 5 Billion Mobile Connections

Cellphone adoption has spread faster than access to basic needs like fresh water or toilets



| Data Table                                                                                            |                                                                                   |                                                                                   |                                                                                    |
|-------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| Era/Device                                                                                            | 1956 IBM Disk Drive                                                               | 1991 Space Shuttle                                                                | Modern Smartphone                                                                  |
|  1956 IBM Disk Drive |  |  |  |
|  Processing Speed    | N/A                                                                               | 1 Megahertz                                                                       | Several Gigahertz                                                                  |
|  Storage Capacity    | 5 Megabytes (1 photo)                                                             | N/A                                                                               | 1 Terabyte (250 HD hours)                                                          |

## Life-Altering Applications



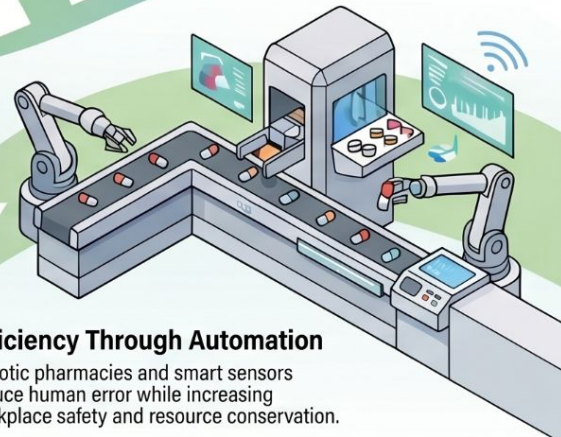
### Restoring Independence for the Disabled

Brain-computer interfaces and smart prosthetics allow paralyzed individuals to operate devices and walk again.



### Efficiency Through Automation

Robotic pharmacies and smart sensors reduce human error while increasing workplace safety and resource conservation.



**Thank You**